

GEOGRAPHY 02 — THE EARTH’S CRUST, ROCKS / INDIA GEOLOGICAL STRUCTURE

CRUST, LITHOSPHERE AND THE → IGNEOUS, SEDIMENTARY AND METAMORPHIC → FROM CONTINENTAL DRIFT TO → INDIAN PLATE JOURNEY AND → INDIAN ROCK SYSTEMS, BASINS → HIMALAYA, FORELAND, COASTS, ISLANDS

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure: the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW

SUBORDINATE EXTRA

APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g11 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

CRUST, LITHOSPHERE AND THE ROCK-CYCLE NETWORK

CRUST, LITHOSPHERE AND THE ROCK-CYCLE NETWORK

EARTH MATERIAL SYSTEM MINERAL

COMPOSITIONAL SHELL

CRUST + RIGID UPPERMOST MANTLE

WEAKER MECHANICAL LAYER BELOW

ROCK-CYCLE NETWORK MAGMA →CRYSTALLISATION→ IGNEOUS WEATHERING + TRANSPORT

<--- MELTING → METAMORPHIC <-- SEDIMENTARY HEAT + PRESSURE

THE CYCLE IS A NETWORK

EARTH MATERIAL SYSTEM mineral → rock → lithology → petrology → geology

Crust = compositional shell

Lithosphere = crust + rigid uppermost mantle

Asthenosphere = weaker mechanical layer below

ROCK-CYCLE NETWORK magma →crystallisation→ IGNEOUS weathering + transport

DEFINITION / COMPONENTS

EARTH MATERIAL SYSTEM mineral → rock → lithology → petrology → geology

Crust = compositional shell

Lithosphere = crust + rigid uppermost mantle

TRIGGER / MECHANISM

Asthenosphere = weaker mechanical layer below

ROCK-CYCLE NETWORK magma →crystallisation→ IGNEOUS weathering + transport

<---- melting → METAMORPHIC <-- SEDIMENTARY heat + pressure compaction + cementation uplift and exposure reconnect all paths

EFFECT / INTERVENTION

The cycle is a network: no rock family has one compulsory next stage.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: The crust is Earth’s outer compositional shell, whereas the lithosphere is the stronger mechanical layer comprising the crust plus the rigid uppermost mantle.

01

STAGE 01

IGNEOUS, SEDIMENTARY AND METAMORPHIC DIAGNOSTIC MATRIX

IGNEOUS, SEDIMENTARY AND METAMORPHIC DIAGNOSTIC MATRIX

FIELD CLUES

MAGMA/LAVA COOLING

CRYSTALS, GLASS, VESICLES INTRUSIVE

SLOW COOLING

COARSE TEXTURE EXTRUSIVE

FAST COOLING

FINE OR GLASSY TEXTURE

FAMILY	PROCESS	FIELD CLUES
Sedimentary	deposit + lithify	beds, fossils, ripples
granite-rhyolite = felsic	diorite-andesite = intermediate gabbro-basalt = mafic	
FIREWALLS bedding is not foliation	loess and moraine are deposits until lithified porosity is void space	permeability is connected flow capacity

CLASSIFICATION

FIELD CLUES

magma/lava cooling

crystals, glass, vesicles intrusive

slow cooling

coarse texture extrusive

DIAGNOSTIC BASIS

fast cooling

fine or glassy texture

deposit + lithify

beds, fossils, ripples

solid-state change

PROCESS LINK

grade, new minerals foliated

differential stress

aligned fabric

Composition pairs

granite-rhyolite = felsic

EXAM TRAP

diorite-andesite = intermediate gabbro-basalt = mafic

FIREWALLS bedding is not foliation

loess and moraine are deposits until lithified porosity is void space

permeability is connected flow capacity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Igneous rocks record the cooling history and composition of magma or lava: slow intrusive cooling favours coarse crystals, while rapid extrusive cooling favours fine, glassy or vesicular textures.

02

STAGE 02

FROM CONTINENTAL DRIFT TO PLATE TECTONICS AND WILSON-CYCLE CLOSURE

FROM CONTINENTAL DRIFT TO PLATE TECTONICS AND WILSON-CYCLE CLOSURE

WEGENER’S DRIFT CLAIM

SHELF-EDGE FIT

MATCHING ROCKS AND FOSSILS

LATER PALAEOMAGNETIC SUPPORT LIMITATION

NO ADEQUATE DRIVING MECHANISM

SEA-FLOOR SPREADING RIDGE CREATION

MAGNETIC STRIPES

WEGENER’S DRIFT CLAIM

shelf-edge fit → matching rocks and fossils

palaeoclimate → later palaeomagnetic support limitation: no adequate driving mechanism

SEA-FLOOR SPREADING ridge creation → magnetic stripes → older/deeper floor outward trench destruction

DEFINITION / COMPONENTS

WEGENER’S DRIFT CLAIM

shelf-edge fit → matching rocks and fossils

palaeoclimate → later palaeomagnetic support limitation: no adequate driving mechanism

TRIGGER / MECHANISM

SEA-FLOOR SPREADING ridge creation → magnetic stripes → older/deeper floor outward trench destruction

PLATE TECTONICS divergence subduction transform creates crust destroys oceanic conserves crust ridge/rift trench + arc shallow quakes rift

narrow sea → ocean → subduction → collision → suture

EFFECT / INTERVENTION

Wilson cycle is an analogy, not a compulsory destiny for every ocean.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plate boundaries are diagnosed by motion and products: divergence creates crust, subduction destroys oceanic crust, continent collision thickens crust, and transform motion conserves crust while generating shallow earthquakes.

03

STAGE 03

INDIAN PLATE JOURNEY AND THE THREE-PART GEOLOGICAL SKELETON

INDIAN PLATE JOURNEY AND THE THREE-PART GEOLOGICAL SKELETON

GONDWANA RIFTING

NORTHWARD INDIAN PLATE DRIFT

TETHYS CLOSURE

INDIA-EURASIA COLLISION ANCIENT PENINSULA HIMALAYAN FOLD-THRUST BELT FORELAND BASIN

NORTH-SOUTH EXAM CROSS-SECTION

SIWALIK HIMALAYA

FORELAND ALLUVIUM

Gondwana rifting → northward Indian Plate drift → Tethys closure

India-Eurasia collision ancient Peninsula Himalayan fold-thrust belt foreland basin cratons + belts shortening + thickening

stable framework continuing uplift/seismicity Indo-Ganga-Brahmaputra

NORTH-SOUTH EXAM CROSS-SECTION

Tethyan / Greater / Lesser / Siwalik Himalaya → foreland alluvium

SYSTEM / DEFINITION

Gondwana rifting → northward Indian Plate drift → Tethys closure

India-Eurasia collision ancient Peninsula Himalayan fold-thrust belt foreland basin cratons + belts shortening + thickening flexure + sediment

stable framework continuing uplift/seismicity Indo-Ganga-Brahmaputra

PROCESS / MECHANISM

NORTH-SOUTH EXAM CROSS-SECTION

Tethyan / Greater / Lesser / Siwalik Himalaya → foreland alluvium

Peninsular cratons, rifts, basins and Deccan cover

SPATIAL / INDIA PATTERN

EVIDENCE LADDER superposition → cross-cutting → unconformity → fossil correlation

radiometric event age → palaeomagnetism → geophysics

Stable Peninsula means relatively stable, not uniform, flat or aseismic.

HAZARD / LIMIT / RESPONSE

NORTH-SOUTH EXAM CROSS-SECTION

MECHANISM / VERDICT — Stable Peninsula means relatively stable, not uniform, flat or aseismic.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India’s geological skeleton consists of an ancient Peninsular shield, an active Himalayan fold-thrust belt and a sediment-filled Indo-Ganga-Brahmaputra foreland basin linked by collision, flexure, erosion and deposition.

04

STAGE 04

INDIAN ROCK SYSTEMS, BASINS AND RESOURCE ASSOCIATIONS

INDIAN ROCK SYSTEMS, BASINS AND RESOURCE ASSOCIATIONS

INDIAN EXPRESSION

SAFE ASSOCIATION

GNEISS + SUPRACRUSTALS

PROCESS-SPECIFIC ORE

CUDDAPAH, VINDHYAN

STONE, BOUNDED ORES

FAULTED INLAND BASINS

INDIAN EXPRESSION

SAFE ASSOCIATION

gneiss + supracrustals

process-specific ore

Cuddapah, Vindhyan

CLASSIFICATION

INDIAN EXPRESSION

SAFE ASSOCIATION

gneiss + supracrustals

process-specific ore

Cuddapah, Vindhyan

DIAGNOSTIC BASIS

stone; bounded ores

faulted inland basins

coal, not universal

Marine basins

coasts and offshore

PROCESS LINK

petroleum potential

Deccan Trap

stacked flood basalts

soil, stone, water

GONDWANA BASINS: Damodar

EXAM TRAP

PETROLEUM SYSTEM

source → maturation → migration → reservoir → seal → trap → timing

DECCAN ARCHITECTURE

flow top / vesicles / fractures / weathered contact / dense flow interior

Association is not determinism: grade, preservation, exploration, technology, economics and environmental limits decide usable resources.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: source → maturation → migration → reservoir → seal → trap → timing.

05

STAGE 05

HIMALAYA, FORELAND, COASTS, ISLANDS AND STRUCTURAL CONTROLS

HIMALAYA, FORELAND, COASTS, ISLANDS AND STRUCTURAL CONTROLS

HIMALAYA COLLISION OROGEN

SUTURE FRAMEWORK

FORELAND FLEXURE AND FILL

BHANGAR/KHADAR ALLUVIAL EXPRESSIONS

PENINSULAR STRUCTURES

COASTS AND ISLANDS

NARMADA-SON AND TAPI RIFTS

FORELAND FLEXURE AND FILL

Himalaya collision orogen: HFT/MFT → MBT → MCT → suture framework

Bhabar → Terai → bhangar/khadar alluvial expressions

PENINSULAR STRUCTURES

SYSTEM / DEFINITION

HIMALAYA collision orogen: HFT/MFT → MBT → MCT → suture framework

FORELAND FLEXURE AND FILL

Bhabar → Terai → bhangar/khadar alluvial expressions

PENINSULAR STRUCTURES

PROCESS / MECHANISM

COASTS AND ISLANDS

Narmada-Son and Tapi rifts

west: rifted-margin sectors

Cambay and Kachchh

SPATIAL / INDIA PATTERN

east: deltaic sediment basins

Godavari rift basin

Andaman: active island arc joints, faults, bedding

Lakshadweep: coral atolls structure guides relief, drainage, waterfalls and groundwater climate + erosion + base level + human

HAZARD / LIMIT / RESPONSE

action modify expression

Rift = extensional zone

graben = down-dropped block

Lineament = mapped linear feature, not automatically an active fault.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: A rift is an extensional crustal zone, a graben is a down-dropped fault block, and a lineament is a mapped linear feature that is not automatically an active fault.

06

STAGE 06

INDIA SEISMIC EXPRESSION AND GEOLOGY-TO-RISK TRANSLATION

INDIA SEISMIC EXPRESSION AND GEOLOGY-TO-RISK TRANSLATION

TECTONIC AND INHERITED STRUCTURE

NE ANDAMAN ARC KACHCHH

PENINSULA ACTIVE COLLISION SUBDUCTION INHERITED FAULTS/RIFTS STRONG SHAKING QUAKE

SITE FILTER

ALLUVIUM, BASIN DEPTH, GROUNDWATER, SLOPE, CONSTRUCTION

EXPOSURE + VULNERABILITY - CAPACITY

DISASTER RISK

TECTONIC AND INHERITED STRUCTURE

Himalaya / NE Andaman arc Kachchh / Peninsula active collision subduction inherited faults/rifts strong shaking

intraplate reactivation

SITE FILTER: alluvium, basin depth, groundwater, slope, construction

SYSTEM / DEFINITION

TECTONIC AND INHERITED STRUCTURE

Himalaya / NE Andaman arc Kachchh / Peninsula active collision subduction inherited faults/rifts strong shaking quake + tsunami

intraplate reactivation

PROCESS / MECHANISM

SITE FILTER: alluvium, basin depth, groundwater, slope, construction

EXPOSURE + VULNERABILITY - CAPACITY

DISASTER RISK

SPATIAL / INDIA PATTERN

BIS Zones II-V are broad design generalisations, not predictions.

Geological association must be followed by site evidence and engineering.

RESPONSE: map → investigate → code → inspect → retrofit → prepare.

HAZARD / LIMIT / RESPONSE

SITE FILTER: alluvium, basin depth, groundwater, slope, construction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India’s seismicity reflects active Himalayan and Andaman plate boundaries plus reactivation of inherited structures such as Kachchh and selected Peninsular faults; seismic zoning is a design generalisation, not earthquake prediction.

07

STAGE 07

MAP-ANSWER AND MAINS SPINE FOR INDIA'S GEOLOGICAL ARCHITECTURE

MAP-ANSWER AND MAINS SPINE FOR INDIA'S GEOLOGICAL ARCHITECTURE

HOW DOES GEOLOGICAL STRUCTURE SHAPE INDIA'S SURFACE GEOGRAPHY?

1. DEFINE

AGE + COMPOSITION + ROCK ARCHITECTURE + TECTONIC STRUCTURES

2. DRAW

FORELAND PLAIN

PENINSULAR SHIELD CROSS-SECTION

3. SEQUENCE

QUESTION: How does geological structure shape India's surface geography?

1. DEFINE: age + composition + rock architecture + tectonic structures

2. DRAW: Himalaya → foreland plain → Peninsular shield cross-section

3. SEQUENCE: Gondwana rift → drift → collision → uplift → sediment fill

4. MAP NAMED UNITS cratons

SYSTEM / DEFINITION

QUESTION: How does geological structure shape India's surface geography?

1. DEFINE: age + composition + rock architecture + tectonic structures

2. DRAW: Himalaya → foreland plain → Peninsular shield cross-section

PROCESS / MECHANISM

3. SEQUENCE: Gondwana rift → drift → collision → uplift → sediment fill

4. MAP NAMED UNITS cratons

Purana basins

SPATIAL / INDIA PATTERN

Gondwana basins

5. LINK PROCESS TO CONSEQUENCE structure → relief/drainage/water/resource/seismic expression

6. QUALIFY association is not reserve; stable is not aseismic; alluvium has structure

HAZARD / LIMIT / RESPONSE

7. VERDICT inherited geology is filtered through tectonics, climate, erosion and use

Answer order: thesis → map → evidence → mechanism → India example

limitation → graded conclusion.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 7. VERDICT inherited geology is filtered through tectonics, climate, erosion and use.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

47. OLDER INDIAN ROCK-SYSTEM SCHEMES

USEFUL ONLY WHEN DECLARED

48. FIELD IDENTIFICATION AND METAMORPHIC NUANCE

49. CRATONS, MOBILE BELTS AND ORE SYSTEMS

50. BASIN ARCHITECTURE BEYOND THE NAME LIST

51. DECCAN HYDROGEOLOGY AND LANDSCAPE REFINEMENT

52. HIMALAYAN STRUCTURAL COMPLEXITY BEYOND A CLASSROOM SECTION

OPTIONAL PROCESS DEPTH

47. Older Indian rock-system schemes: useful only when declared

48. Field identification and metamorphic nuance

49. Cratons, mobile belts and ore systems

50. Basin architecture beyond the name list

DATA / INSTITUTION LIMIT

51. Deccan hydrogeology and landscape refinement

52. Himalayan structural complexity beyond a classroom section

53. Foreland, sediment routing and basin response

54. Hard-rock aquifers and engineering investigations

CONTEMPORARY PARALLEL

55. Mineral security without geological overclaim

56. Map and answer architecture

Granite, diorite and gabbro provide coarse intrusive end-members; rhyolite, andesite and basalt are broad extrusive partners.

Slate cleavage, schistosity and gneissic banding are fabrics, not sedimentary beds.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.